Toward Vision-Guided Pick-and-Place for Tabletop Humanoid Manipulation

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Outline

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Motivation

- Many robotic systems work well only in structured settings with predefined objects and tasks, rather than in open-ended human environments [KBP13, KMvB+21].
- Real-world environments are less stable: layouts change, new objects appear, and uncertainty affects both perception and control [KMvB+21, ZYS+25].
- For humanoid robots, a key capability is **perception-to-manipulation**: detecting an object, estimating how to grasp it, and executing manipulation reliably [ZYS+25].
- This project studies that capability in a controlled tabletop setting using the Fourier GR-1 humanoid as a first step toward more adaptive embodied behavior.

Problem Setting

Long-term vision

Build an integrated visuomotor system in which the robot:

1. observes a scene,
2. identifies a target object,
3. estimates the object information needed for grasping,
4. executes grasp-and-place behavior robustly.

- Current focus: controlled tabletop pick-and-place.
- Immediate goal: establish reliable manipulation on hardware.
- Longer-term goal: integrate this manipulation stack with a perception module that can adapt to new objects over time [KMvB+21, ZWY+24].

Why It Is Challenging

- **Sim-to-real gap:** policies trained in simulation often degrade on physical hardware [TFR+17].
- **Manipulation sensitivity:** contact-rich behaviors are highly affected by geometry, friction, and alignment error [TFR+17, ZYS+25].
- **Pipeline fragility:** if perception shifts or object pose changes, downstream manipulation can fail [KMvB+21, ZYS+25].
- **Over-engineering risk:** scripted systems often depend on heuristic waypoints, thresholds, and fixed assumptions.

Visuomotor Manipulation

Theme	What Prior Work Shows	Limitation
CLIPort [SMF22]	Semantic grounding with spatial manipulation	Still assumes fixed perception during task execution
PerAct [SMF23]	Structured 3D action representations for multi-task manipulation	Mostly evaluated under fixed object categories and benchmark settings
RT-1 / Octo / Diffusion Policy [BBC+23, GWP+24, CXF+25]	Large-scale policy learning improves robustness and generalization	Continual adaptation after deployment is not central

Takeaway: Visuomotor manipulation is strong, but it still largely relies on fixed perception and fixed deployment distributions.

Continual Perception and Adaptation

Area	What Exists	What Is Missing
LwF / iCaRL [LH16, RKSL17]	Mitigate forgetting during incremental learning	Mostly validated on vision benchmarks, not robot behavior
LoRA [HSW+22]	Efficient adaptation through lightweight modules	Efficient updating does not guarantee stable manipulation
Lifelong robotics [KMvB+21, MBY+25]	Continuous learning is required for long-term autonomy	Coupling adaptive perception and action remains limited

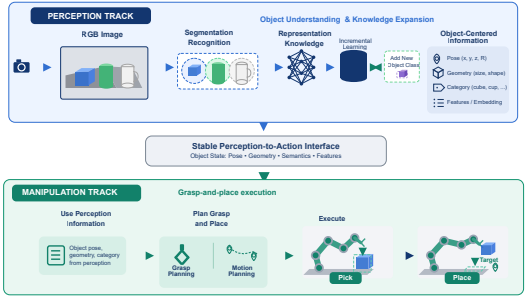
Takeaway: Continual learning supports perception, but embodied deployment remains much less developed.

Research Gap

Theme	What Prior Work Shows	Limitation
Manipulation systems [SMF23, BBC+23, GWP+24]	Strong task execution and policy learning	Usually assumes fixed perception and stable object knowledge
Continual perception [HSW+22, RKSL17, ZWY+24]	Incremental knowledge update with reduced forgetting	Rarely evaluated through downstream grasping or placement
Embodied adaptation [KMvB+21, MBY+25]	Calls for perception and action to evolve together	A stable perception-to-action interface is still missing

Takeaway: The key gap is stable coordination between evolving perception and downstream action.

End-to-End System Diagram



The proposed system connects perception and manipulation through a shared perception-to-action interface.

Methodology: Key Idea

- The long-term target is a modular visuomotor pipeline.
- The **perception module** estimates object-centered information and expands object knowledge over time.
- The **manipulation module** uses that information to generate grasp-and-place behavior.
- In this work, these two parts are studied separately through a perception study and a manipulation study.
- The long-term goal is to connect them through a stable perception-to-action interface [ZYS+25, KMvB+21].

Manipulation Objective

Goal

Develop a tabletop pick-and-place capability for the Fourier GR-1 humanoid in simulation.

- Initial assumption: cube position is available from simulation state.
- Task: grasp a cube from the table and place it at a predefined target location.
- Main question: how far can a scripted controller go, and where does learned control become necessary?

Simulation Setup

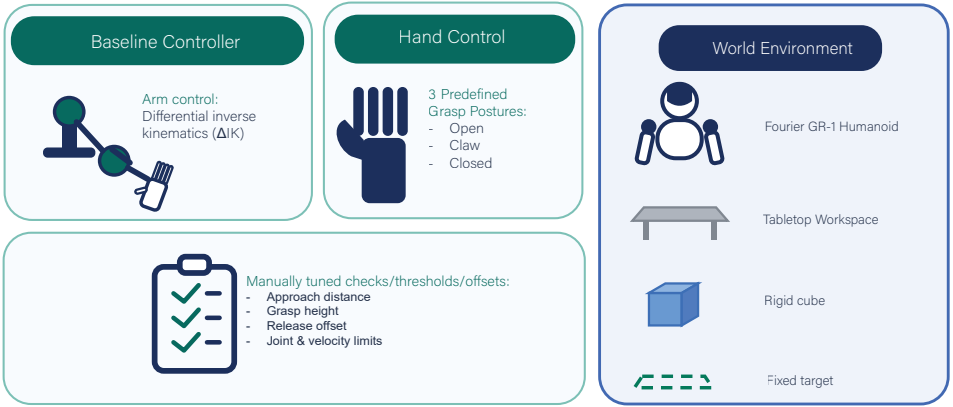


Figure: Manipulation task environment

Scripted Baseline

- The first controller is a **hand-designed state machine**.
- This provides a structured and debuggable baseline for tabletop pick-and-place.

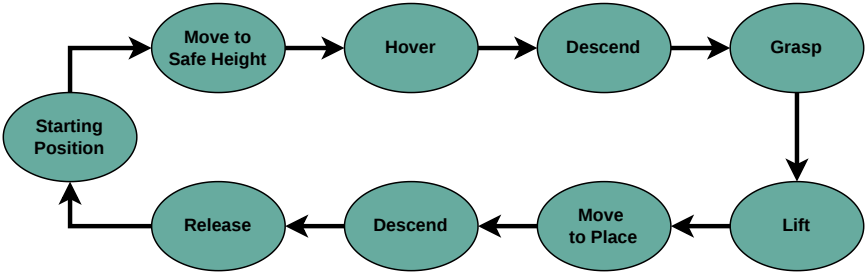


Figure: Stage flow diagram

Baseline Control Structure

- **Arm control:** differential inverse kinematics maps desired end-effector pose to arm joint targets.
- **Hand control:** predefined hand postures (*open*, *claw*, *closed*).
- **State transitions:** based on manually designed conditions:
  - XY alignment,
  - Z alignment,
  - hand-to-cube proximity,
  - lift height,
  - hold counters.

Assumptions and Constraints

- Cube position is directly available from simulator state.
- Placement target is predefined.
- End-effector orientation is fixed.
- Motion depends on manually tuned:
  - height offsets,
  - palm alignment biases,
  - stage thresholds,
  - grasp posture schedules.
- Cube pose is randomized slightly across episodes, but the control strategy remains scripted.

What the Baseline Provides

Value of the baseline

It shows that the robot can perform tabletop pick-and-place under controlled assumptions.

- Establishes the scene, robot, and control pipeline.
- Makes failure modes easier to inspect.
- Provides a reference system before learning-based control.

Main limitation

The system still depends on heuristic offsets, thresholds, and predefined stage logic.

RL Reformulation

- To reduce dependence on hand-crafted stage logic, the task was reformulated as a **reinforcement learning problem**.
- A parallel Isaac Lab environment was created for multi-environment training.
- Training uses **PPO**.
- Goal: allow the robot to discover grasping and transport behavior through interaction rather than explicit stage scripting [KBP13].

RL Environment

- **Observation includes:**
  - end-effector position,
  - cube position,
  - placement target position,
  - relative vectors,
  - arm joint positions,
  - hand joint positions.
- **Action includes:**
  - incremental arm joint commands,
  - one scalar grip-control signal.
- Cube position is randomized at reset to avoid overfitting to a single trajectory, following the general motivation of reducing sim-specific overfitting [TFR+17].

Reward Design

- Reward encourages:
  1. reaching the cube,
  2. lifting the cube,
  3. transporting it toward the goal,
  4. successful placement.
- A penalty on action magnitude discourages unnecessarily aggressive control.
- Episodes terminate on:
  - success,
  - cube drop,
  - out-of-bounds behavior,
  - timeout.

Interpretation

- The manipulation work follows a clear progression:
  1. establish a scripted baseline,
  2. identify brittle hand-crafted assumptions,
  3. reformulate the task as learned control.
- This study does not yet solve fully dynamic grasping.
- It does establish the control-side foundation required for a future integrated visuomotor pipeline [ZYS+25].

Perception Objective

Goal

Develop an incremental semantic segmentation module that can learn new object classes over time while minimizing catastrophic forgetting.

- Motivation: in open-world robotics, new objects appear after deployment [KMvB+21].
- Full retraining is slow and impractical for continual operation.
- The perception module should expand object knowledge while preserving old classes [ZWY+24].
- In the long term, this module will provide object-centered information for downstream grasping and placement.

Perception Flow Diagram

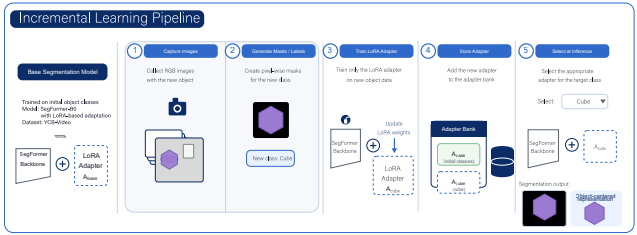


Figure: Perception incremental learning pipeline

Proposed Pipeline

- Start with a base segmentation model trained on initial object classes.
- When a robot encounters a new object:
  1. capture images,
  2. generate labels and masks automatically,
  3. train a new LoRA adapter,
  4. store the adapter in a library,
  5. select the correct adapter at inference.
- Long-term goal: continual perception without destroying prior knowledge [HSW+22, ZWY+24].

Current scope

In this work, the full online robot pipeline was not implemented, so the study focuses on offline incremental segmentation experiments.

Dataset and Setup

- **Dataset:** YCB-Video
  - 21 household object classes,
  - RGB-D video with pixel-level annotations,
  - class split: base classes 1–10, new classes 11–21.
- **Proof-of-concept subset:**
  - 5 sequences,
  - 6,638 usable frames,
  - 5,974 train / 664 test.
- **Model:** SegFormer-B0 with modified segmentation head.
- **Adaptation:** LoRA inserted into selected transformer layers [HSW+22].
- **Metric:** mean Intersection-over-Union (mIoU).

Incremental Learning Results

Method	Base IoU Before	Base IoU After	New Class IoU
Naive Fine-Tuning	0.96	0.18	0.95
LoRA Fine-Tuning	0.96	0.19	0.93
Adapter Fusion	0.96	0.97	0.42
LoRA + Distillation	0.96	0.8833	0.9144

- Naive and plain LoRA fine-tuning learned new classes but forgot old ones, consistent with catastrophic forgetting in sequential learning [MC89, Fre99].
- Adapter fusion preserved old knowledge but underfit the new classes.
- **LoRA + knowledge distillation gave the best balance** between retention and new-class learning, matching the intuition behind LwF-style retention mechanisms [LH16, HSW+22].

Manipulation Progression

Domain	Capability	Level 0 *(Hard-coded)*	Level 1 *(Reactive)*	Level 2 *(Adaptive)*	Level 3 *(Autonomous)*	Level 4 *(Dynamic)*
Manipulation	**Grasp Planning**	None	Model-based (Stage 2)	Model-free isolated (Stage 3)	Model-free cluttered	Articulated / deformable
	Execution	Hard-coded (Stage 1)	Reactive single-modal	Reactive multi-modal (Stage 3)	Autonomous under uncertainty (Target)	Autonomous dynamic

Reading: The current system moves from a hard-coded baseline toward a model-free RL policy as the next major capability step.

Perception Progression

Domain	Capability	Level 0 *(Static)*	Level 1 *(Offline Incremental)*	Level 2 *(Retention-Aware)*	Level 3 *(Online Adaptive)*	Level 4 *(Embodied Continual)*
Perception	**Object Knowledge**	Fixed class set	Offline class-incremental updates	LoRA + distillation retention	Online novel-class expansion	Continual open-world object learning
	System Integration	Standalone segmentation	Dataset-level evaluation	Incremental method comparison	On-robot perception updates	Perception-action integration (Target)

Reading: The current perception system has moved beyond static segmentation into offline incremental learning, with LoRA + distillation as the strongest retention-aware baseline.

Next Steps

- On the perception side, move from offline incremental segmentation toward online novel-class adaptation and robot-level integration [KMvB+21, MBY+25].
- On the manipulation side, improve the RL policy until grasping is more robust to pose variation.
- Reduce reliance on fixed offsets and predefined stage assumptions in manipulation.
- Integrate the perception and manipulation modules into a single visuomotor pipeline.
- After stable simulation performance, transfer the integrated system to the physical Fourier GR-1 robot.

Conclusion

- This work targets a foundational robotics problem: tabletop perception-to-manipulation with a humanoid robot [ZYS+25].
- The current contribution on the manipulation side is a progression from scripted control to RL-based adaptive control.
- The perception-side contribution is an incremental segmentation study showing that LoRA + distillation gives the strongest retention/plasticity trade-off in the current setup [LH16, HSW+22].
- Together, these studies establish the basis for a larger integrated visuomotor system.

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Thank You

Questions?